

torch.functional.unravel_index#

https://pytorch.org/docs/stable/generated/torch.functional.unravel_index.html

Description:

Converts a tensor of flat indices into a tuple of coordinate tensors that index into an arbitrary tensor of the specified shape. **indices** (Tensor) – An integer tensor containing indices into the flattened version of an arbitrary tensor of shape **shape**. All elements must be in the range $[0, \text{prod}(\text{shape}) - 1]$. **shape** (int, sequence of ints, or torch.Size) – The shape of the arbitrary tensor. All elements must be non-negative. Each *i*-th tensor in the output corresponds with dimension *i* of **shape**. Each tensor has the same shape as **indices** and contains one index into dimension *i* for each of the flat indices given by **indices**.

Function Signature

```
torch.functional.unravel_index(indices, shape)[source]#
```

Parameters

Returns

Return type

Returns:

tuple of Tensors

Code Examples

```
>>> import torch
>>> torch.unravel_index(torch.tensor(4), (3, 2))
(tensor(2),
 tensor(0))

>>> torch.unravel_index(torch.tensor([4, 1]), (3, 2))
(tensor([2, 0]),
 tensor([0, 1]))

>>> torch.unravel_index(torch.tensor([0, 1, 2, 3, 4, 5]), (3,
2))
(tensor([0, 0, 1, 1, 2, 2]),
 tensor([0, 1, 0, 1, 0, 1]))

>>> torch.unravel_index(torch.tensor([1234, 5678]), (10, 10, 10,
10))
(tensor([1, 5]),
 tensor([2, 6]),
 tensor([3, 7]),
 tensor([4, 8]))

>>> torch.unravel_index(torch.tensor([[1234], [5678]]), (10, 10,
10, 10))
(tensor([[1], [5]]),
 tensor([[2], [6]]),
 tensor([[3], [7]]),
 tensor([[4], [8]]))

>>> torch.unravel_index(torch.tensor([[1234], [5678]]), (100,
100))
(tensor([[12], [56]]),
 tensor([[34], [78]]))
```

Detailed Documentation

Documentation

Converts a tensor of flat indices into a tuple of coordinate tensors that index into an arbitrary tensor of the specified shape.

`indices` (Tensor) – An integer tensor containing indices into the flattened version of an arbitrary tensor of shape `shape`. All elements must be in the range $[0, \text{prod}(\text{shape}) - 1]$.

`shape` (int, sequence of ints, or `torch.Size`) – The shape of the arbitrary tensor. All elements must be non-negative.

Each i -th tensor in the output corresponds with dimension i of `shape`. Each tensor has the same shape as `indices` and contains one index into dimension i for each of the flat indices given by `indices`.